

Lifelong Learning Techniques

for an Origami Robot

by

E. De Vroey

Student number: 4685156

Contents

1	Introduction	1
1.1	Application to soft (origami) robots	1
1.2	Challenges	1
2	Background and definition	3
3	Blackbox methods	5
3.1	Learning control policies	5
3.2	Learning the dynamic model	5
3.3	Incremental learning.	6
4	Learning architectures	9
4.1	Dual architectures	9
4.2	Architectural expansions	9
5	Storage and retrieval	11
6	Discussion and conclusion	13

1

Introduction

1.1. Application to soft (origami) robots

In the field of soft robotics, lifelong learning are particularly interesting.

1.2. Challenges

(Kim et al., 2021; Lesort et al., 2020).

How can lifelong learning methods be employed to manage the evolving dynamics due to wear and tear in the control of a soft origami robot over its operational lifespan?

2

Background and definition

(Annaswamy & Fradkov, 2021; Landau et al., 2011; Lesort et al., 2020).

Blackbox methods

general intro, define "blackbox"

(Alessi et al., 2023; Gijsberts & Metta, 2011; Nguyen-Tuong et al., 2009; Romeres et al., 2016; Thuruthel et al., 2019; Zhou et al., 2022).

3.1. Learning control policies

A perfect example of a blackbox method is given by Alessi et al. (2023). In their study they used proximal policy optimization (Schulman et al., 2017) to obtain a controller that can generalize over new dynamics and tasks. The controller is a blackbox: a desired trajectory, several measurements of the robots state, and other values of interest, such as the error of robot's end-effector tip, are used as inputs, resulting in a control action. It essentially acts in a manner akin to a feedback controller using an inverse dynamics model, only in this case it is unknown what is happening *under the hood*.

The previous example is "*perfect*" in the sense that it perfectly embodies the simplistic nature of blackbox methods, however it is worth noting that other techniques, such as more complicated architectures or combinations of multiple models, can be used in combination with these blackbox methods. Such architectural modifications will be discussed in chapter 4.

3.2. Learning the dynamic model

One way of having continually adapting control for soft robots, is by implementing a (traditional) control scheme that uses a dynamic model (either forward or inverse) of the robot and then to incrementally update that model. The idea of learning a dynamic model for soft robots has advantages even outside the context of lifelong adaptation: many soft robots have such a large amount of degrees of freedom and such complex and nonlinear dynamics, that analytically deriving a dynamic model is both cumbersome and often insufficiently accurate (Gillespie et al., 2018; Giorelli et al., 2015; Kim et al., 2021; Thuruthel et al., 2019).

Blackbox methods also allow for learning both forward and inverse dynamic models and can thus be used in several control schemes. When learning a forward model, the robot's actions and resulting states can be used as the inputs and outputs, respectively, this is so-called *direct modeling* (Nguyen-Tuong & Peters, 2011b). In the case of learning inverse dynamic models, such direct approaches might not always be applicable, due to the multiple solutions that exist for the inverse dynamics of robots with redundant degrees of freedom. Instead, an *indirect-modeling* approach can be taken, where the model is trained to minimize the error of the feedback controller (Nguyen-Tuong & Peters, 2011b).

The form that the dynamic model takes can vary. For example, in Gillespie et al. (2018), a neural network is trained to predict the next state of the robot, given the current state and a

control action, the analytical gradients of these outputs w.r.t. to the inputs are made available during the process of backpropagation and can be used in matrices A and B to form a more interpretable state-space system. In Thuruthel et al. (2019) on the other hand, a forward model is learned for open-loop control, using the control action, the current state, and the previous state as input, and the next state as output to train a recurrent neural network. This forward model is kept in its more abstract blackbox form, and is directly used to generate trajectories in open-loop control.

Likewise, since the lifelong adapting nature of the techniques is embedded in the dynamic model, the choice of control scheme in which the learned dynamic model is used can vary from implementation to implementation. To illustrate, the aforementioned matrices A and B from Gillespie et al. (2018) are used in a model predictive control setting, while the generated trajectories from Thuruthel et al. (2019) are used as samples to train a blackbox control policy (the dynamic model is in essence an intermediary step in order to create a blackbox policy as in section 3.1).

Contrary to blackbox policy learning methods, the adaptation to new tasks is not included in the learning process of the dynamic model, but is entirely dependent on the control policy. In case control policies have been defined in advance for several tasks, these task settings might serve as a test to verify that the dynamic model generalizes well over these different settings. In situations where tasks are not known in advance, a control scheme such as an incrementally updated (inverse) dynamics model in some form of a feedback control loop might serve for simple applications (Nguyen-Tuong & Peters, 2011b). For more complex tasks, a more intricate control system might be warranted.

3.3. Incremental learning

In section 3.1 and 3.2 an overview was provided of how blackbox machine learning techniques can be used to model either control policies or dynamic models (to be used in control policies). The choice of machine learning method can be important when it comes to the desired incremental nature of the learning techniques. To illustrate, consider these two main strategies in continually adaptive control:

1. All training is performed offline (after gathering sufficient data) and the robot is now completely capable of immediately adapting to changing dynamics it will encounter during its operational lifespan.
2. Training is performed incrementally as data comes in, which can potentially be done in real-time.

In both scenarios, the control of the robot would be continually adaptive. However, only strategy 2 aligns with the incremental objectives of the adaptive control challenge. In addition, the complex dynamics and many degrees of freedom of soft robots might make it difficult to accurately represent all possible degradations in dynamics (such as damages) or state-space configurations of the robot, potentially causing an imbalance in the training dataset of strategy 1. Incrementally learning to adapt the control policy or dynamic model is thus far more desirable, since it can cope with unforeseen circumstances.

¹For generating training samples

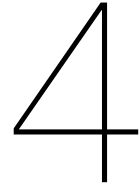
²After training on generated samples

list all other methods that use black-box/interpretable models? instead of just illustrative examples

footnotes, checkmarks, table size, consistency: mention the form (eg. NN, gaussian process, ...) or the training algorithm (eg. PPO)

Reference	Dynamic model	Control policy	Incremental
Gillespie et al. (2018)	SS-matrices	MPC	✓
Thuruthel et al. (2019) ¹	RNN	Open-loop + trajectory optimization	✓
Lu et al. (2020)	Blackbox	MPC	✓
Nguyen-Tuong and Peters (2011a)	GPR	Feedforward	✓
Romeres et al. (2016)	Blackbox	TODO	✓
Nazeer et al. (2023)	Blackbox	Blackbox	✓
Thuruthel et al. (2019) ²	–	NN	✓
Alessi et al. (2023)	–	PPO	–

Table 3.1: An overview blackbox methods and their ability to be implemented incrementally. For dynamic model learning methods, the control policy specifies the control scheme in which the model is used.



Learning architectures

4.1. Dual architectures

(Lu et al., 2019, 2020; Mallya et al., 2018; Schwarz et al., 2018; Shin et al., 2017; Sprechmann et al., 2018).

4.2. Architectural expansions

(Chatzilygeroudis et al., 2016; Gillespie et al., 2018; Nazeer et al., 2023).

5

Storage and retrieval

(Aljundi et al., 2016; Mouret & Clune, 2015; Nguyen-Tuong & Peters, 2011a; Schulman et al., 2017; Shin et al., 2017; Sprechmann et al., 2018; Wang et al., 2022; Xie & Finn, 2021).

6

Discussion and conclusion

Todo list

general intro, define "blackbox"	5
list all other methods that use blackbox/interpretable models? instead of just illustrative examples	6
footnotes, checkmarks, table size, consistency: mention the form (eg. NN, gaussian process, ...) or the training algorithm (eg. PPO)	6

Bibliography

- Alessi, C., Hauser, H., Lucantonio, A., & Falotico, E. (2023). Learning a controller for soft robotic arms and testing its generalization to new observations, dynamics, and tasks. *2023 IEEE International Conference on Soft Robotics (RoboSoft)*, 1–7. <https://doi.org/10.1109/RoboSoft55895.2023.10121988>
- Aljundi, R., Chakravarty, P., & Tuytelaars, T. (2016). Expert gate: Lifelong learning with a network of experts. <http://arxiv.org/abs/1611.06194>
- Annaswamy, A. M., & Fradkov, A. L. (2021, January). A historical perspective of adaptive control and learning. <https://doi.org/10.1016/j.arcontrol.2021.10.014>
- Chatzilygeroudis, K., Cully, A., & Mouret, J.-B. (2016). Towards semi-episodic learning for robot damage recovery. <http://arxiv.org/abs/1610.01407>
- Gijsberts, A., & Metta, G. (2011). Incremental learning of robot dynamics using random features. *2011 IEEE International Conference on Robotics and Automation*, 951–956. <https://doi.org/10.1109/ICRA.2011.5980191>
- Gillespie, M. T., Best, C. M., Townsend, E. C., Wingate, D., & Killpack, M. D. (2018). Learning nonlinear dynamic models of soft robots for model predictive control with neural networks. *2018 IEEE International Conference on Soft Robotics (RoboSoft)*, 39–45. <https://doi.org/10.1109/ROBOSOFT.2018.8404894>
- Giorelli, M., Renda, F., Calisti, M., Arienti, A., Ferri, G., & Laschi, C. (2015). Learning the inverse kinetics of an octopus-like manipulator in three-dimensional space. *Bioinspiration & Biomimetics*, 10, 035006. <https://doi.org/10.1088/1748-3190/10/3/035006>
- Kim, D., Kim, S.-H., Kim, T., Kang, B. B., Lee, M., Park, W., Ku, S., Kim, D., Kwon, J., Lee, H., Bae, J., Park, Y.-L., Cho, K.-J., & Jo, S. (2021). Review of machine learning methods in soft robotics (G. Gu, Ed.). *PLOS ONE*, 16, e0246102. <https://doi.org/10.1371/journal.pone.0246102>
- Landau, I. D., Lozano, R., M'Saad, M., & Karimi, A. (2011). *Adaptive control*. Springer London. <https://doi.org/10.1007/978-0-85729-664-1>
- Lesort, T., Lomonaco, V., Stoian, A., Maltoni, D., Filliat, D., & Díaz-Rodríguez, N. (2020). Continual learning for robotics: Definition, framework, learning strategies, opportunities and challenges. *Information Fusion*, 58, 52–68. <https://doi.org/10.1016/j.inffus.2019.12.004>
- Lu, K., Grover, A., Abbeel, P., & Mordatch, I. (2020). Reset-free lifelong learning with skill-space planning. <http://arxiv.org/abs/2012.03548>
- Lu, K., Mordatch, I., & Abbeel, P. (2019). Adaptive online planning for continual lifelong learning. <http://arxiv.org/abs/1912.01188>
- Mallya, A., Davis, D., & Lazebnik, S. (2018). Piggyback: Adapting a single network to multiple tasks by learning to mask weights. <http://arxiv.org/abs/1801.06519>
- Mouret, J.-B., & Clune, J. (2015). Illuminating search spaces by mapping elites. <http://arxiv.org/abs/1504.04909>
- Nazeer, M. S., Bianchi, D., Campinoti, G., Laschi, C., & Falotico, E. (2023). Policy adaptation using an online regressing network in a soft robotic arm. *2023 IEEE International Conference on Soft Robotics (RoboSoft)*, 1–7. <https://doi.org/10.1109/RoboSoft55895.2023.10121927>

- Nguyen-Tuong, D., & Peters, J. (2011a). Incremental online sparsification for model learning in real-time robot control. *Neurocomputing*, 74, 1859–1867. <https://doi.org/10.1016/j.neucom.2010.06.033>
- Nguyen-Tuong, D., & Peters, J. (2011b). Model learning for robot control: A survey. *Cognitive Processing*, 12, 319–340. <https://doi.org/10.1007/s10339-011-0404-1>
- Nguyen-Tuong, D., Seeger, M., & Peters, J. (2009). Model learning with local gaussian process regression. *Advanced Robotics*, 23, 2015–2034. <https://doi.org/10.1163/016918609X12529286896>
- Romeres, D., Zorzi, M., Camoriano, R., & Chiuso, A. (2016). Online semi-parametric learning for inverse dynamics modeling. <http://arxiv.org/abs/1603.05412>
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. <http://arxiv.org/abs/1707.06347>
- Schwarz, J., Luketina, J., Czarnecki, W. M., Grabska-Barwinska, A., Teh, Y. W., Pascanu, R., & Hadsell, R. (2018). Progress & compress: A scalable framework for continual learning. <http://arxiv.org/abs/1805.06370>
- Shin, H., Lee, J. K., Kim, J., & Kim, J. (2017). Continual learning with deep generative replay. <http://arxiv.org/abs/1705.08690>
- Sprechmann, P., Jayakumar, S. M., Rae, J. W., Pritzel, A., Badia, A. P., Uria, B., Vinyals, O., Hassabis, D., Pascanu, R., & Blundell, C. (2018). Memory-based parameter adaptation. <http://arxiv.org/abs/1802.10542>
- Thuruthel, T. G., Falotico, E., Renda, F., & Laschi, C. (2019). Model-based reinforcement learning for closed-loop dynamic control of soft robotic manipulators. *IEEE Transactions on Robotics*, 35, 124–134. <https://doi.org/10.1109/TRO.2018.2878318>
- Wang, Z., Chen, C., & Dong, D. (2022). Lifelong incremental reinforcement learning with online bayesian inference. *IEEE Transactions on Neural Networks and Learning Systems*, 33, 4003–4016. <https://doi.org/10.1109/TNNLS.2021.3055499>
- Xie, A., & Finn, C. (2021). Lifelong robotic reinforcement learning by retaining experiences. <http://arxiv.org/abs/2109.09180>
- Zhou, W., Bohez, S., Humplik, J., Abdolmaleki, A., Rao, D., Wulfmeier, M., Haarnoja, T., & Heess, N. (2022). Forgetting and imbalance in robot lifelong learning with off-policy data. <http://arxiv.org/abs/2204.05893>